

Manual

piCOBOT® Electric FANUC CRX

Electrical vacuum gripper





This manual is available in the following languages at piab.com

The original manual is written in English.



中文



English



Français



Deutsch



Italiano



日本語



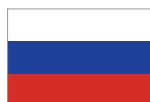
한국어



Polski



Português (Brasil)



Русский



Español



Svenska

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1. Introduction to the manual

1.1. About the manual

- The responsible party for the production site must ensure that this manual will be read and understood.
- The section on safety should be studied extra carefully.
- The manual should be stored in a known and easily accessible place, which can be digital.
- Before doing service and maintenance on the equipment, carefully study the applicable parts of the manual.

This manual (Art no. 0249448, Rev.00, en-GB, 2025-03) is for piCOBOT® Electric FANUC CRX.

1.2. Safety signs used in the manual

Take note of all the warnings, mandatory, and other signs in this manual. They have the following meanings:

1.2.1. Warning signs



Warning

Failure to follow the instructions may result in death or serious injury!



Warning

Vacuum force



Warning

Hot surface

1.2.2. Mandatory signs



Notice

Information that needs extra attention!



Important

Wear eye protection



Important

Wear ear protection

1.3. Target group

This manual, especially the section about safety, shall be read by all staff who will perform any type of work with the product or equipment:

- Installation personnel
- Operating personnel
- Service and maintenance personnel
- Cleaning personnel (cleaning of equipment and the area around it)

1.4. Test conditions

Unless otherwise stated, the values specified in this manual are tested at:

- Room temperature: (20°C [68°F] ± 3°C [5.5°F])
- Standard atmosphere: (101.3 kPa [29.9 inHg] ± 1.0 kPa [0.3 inHg])

2. Safety instructions

2.1. Disclaimer

Piab AB is not responsible for installation and operation of piCOBOT® Electric in a robot system. The required steps must be undertaken in supervision and approval by authorized integrators.

Piab AB is not responsible for safety of a complete robot system with piCOBOT® Electric installed. The required steps must be undertaken in supervision and approval by authorized integrators.

2.2. General safety

The correct use of the product within a system is the responsibility of the system designer or person responsible for the technical specifications.

Vacuum may lead to the implosion of closed containers.

Ensure that components are properly secured; regularly check that connections are in good working order, as high cycles or vibrations may cause them to loosen.

2.3. Safe usage

The product described in this manual is designed for implementation in industrial systems; therefore, it must not be used under conditions other than those specified herein.



Warning

- The use of a damaged product may result in bursting and cause injury or property damage.
- Restricted exhaust may result in decreased product performance and/or application failure.
- Ensure that no foreign matters or objects enter the vacuum port, due to the risk of damage to the product.
- Do NOT use the product for evaporating flammable liquids.
- Do NOT use the product for evacuating liquids.
- Do NOT use the product in potentially explosive environments.



Warning

Vacuum can cause severe injuries. Keep hands, legs, hair, and eyes away from vacuum inlets.

2.3.1. Installation

Prior to assembly and disassembly of the product, turn off the electrical supply. Installing, using, and maintaining the product should solely be carried out by experienced and specially trained personnel.

2.3.2. Operation



Important

Wear eye protection when operating piCOBOT® Electric.



Important

Wear ear protection when operating piCOBOT® Electric.

2.3.3. Maintenance



Warning

Prior to any maintenance work, turn off or disconnect the electrical power supply and ensure that the motor of the pump has cooled down.



Warning

Hot motor surface can cause severe injuries, ensure the pump has cooled down prior to disassembly.

Maintenance must only be carried out after thoroughly reading and understanding this manual.

2.4. Risk assessment

The integrator must perform a risk assessment on the complete robot application. The piCOBOT® Electric is only a component in a robot application thus the safe use of the piCOBOT® Electric relies on the integrator's ability to design a safe robot application. The piCOBOT® Electric is designed with features especially suitable for collaborative applications, such as:

- Compact design to minimize the footprint of the product in limited working spaces.
- Low weight to reduce the impact inertia.
- Large surface area-to-volume ratio, to reduce the impact inertia (pressure).
- Small build height (tool center point close to robot flange) reduces the force exerted on the robot joints during motion.
- Rounded edges.
- Intuitive installation of the pump unit to the robot interface.
- Sufficient number of fasteners for all attachment interfaces to ensure mechanical integrity.
- LED lights on the product for indicating different operational states in different colors that allows for corrective actions being taken, away from the hazardous area.
- Securing the arms of the Adjustable Gripper (optional gripping tool) defines a maximum load that the arms can handle before they are displaced in case of an uncontrolled impact.

2.5. Intended use

- For professional use only.
- The product shall be used in environments within the product's specifications and certifications.
- The product shall be used to evacuate air (non-liquids) from a volume to create vacuum for gripping, holding, and lifting processes.

- The product is intended to be used in on/off operation, turning the vacuum on and off for every operational cycle.
- The piCOBOT® Electric is an end-of-arm tooling on collaborative robots, used for gripping and holding objects. Typical applications are:
 - Pick-and-place
 - Top loading
 - Machine tending
 - Bin picking
 - Order fulfillment
 - Assembly

2.6. Misuse

The product is only cleared for operation under the conditions stated in this manual and in the datasheets. Any use or application deviating from the intended use is deemed to be impermissible misuse. This includes, but is not limited to:

- Use of product that is damaged.
- Use of the product for evacuating liquids.
- Use of electrical voltage outside specification.
- Use of the product in potentially explosive atmospheres.
- Use of the product to evacuate flammable liquids.
- Use of the product running vacuum continuously with more than 60% operational running time per cycle (less than 40% of the time turned off). The product is intended to be operated in an ON and OFF manner.
- Use of the product as a standalone safety unit to fulfil international lifting standards.
- Use of the product to evacuate solid content without the use of a filter.
- Use of the product if the exhaust is restricted or blocked.
- Use of the product in direct contact with water spray or excessive amounts of water.
- Use of the product in potentially explosive atmospheres.
- Use of the product in life-critical applications.

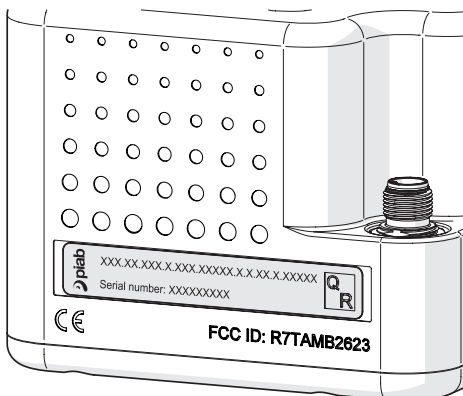
3. Introduction to the piCOBOT® Electric

3.1. Manufacturer

Piab AB
P.O. Box 146
SE-182 12 Danderyd
SWEDEN

3.2. Identification label

Each unit is identified by a label with identification information. For any communication with Piab AB or service centers always refer to the label information and state both the Product code and Serial number.



3.3. Compliance



European Directives, CE

Directive	Standard and/or measurement reference
Electromagnetic Compatibility (EMC) Directive 2014/30/EU	EN/(IEC) 61000-6-2:2019 EN/(IEC) 61000-6-4:2019
RoHS2 Directive (2011/65/EU)	Compliant



Federal Communications Commission, FCC

FCC ID	R7TAMB2623
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3.4. For use with FANUC CRX

piCOBOT® Electric FANUC CRX is intended to be used with FANUC CRX cobots.

It uses RS-485 serial communication and the FANUC CRX Plugin software for operation and control.

3.5. Overview

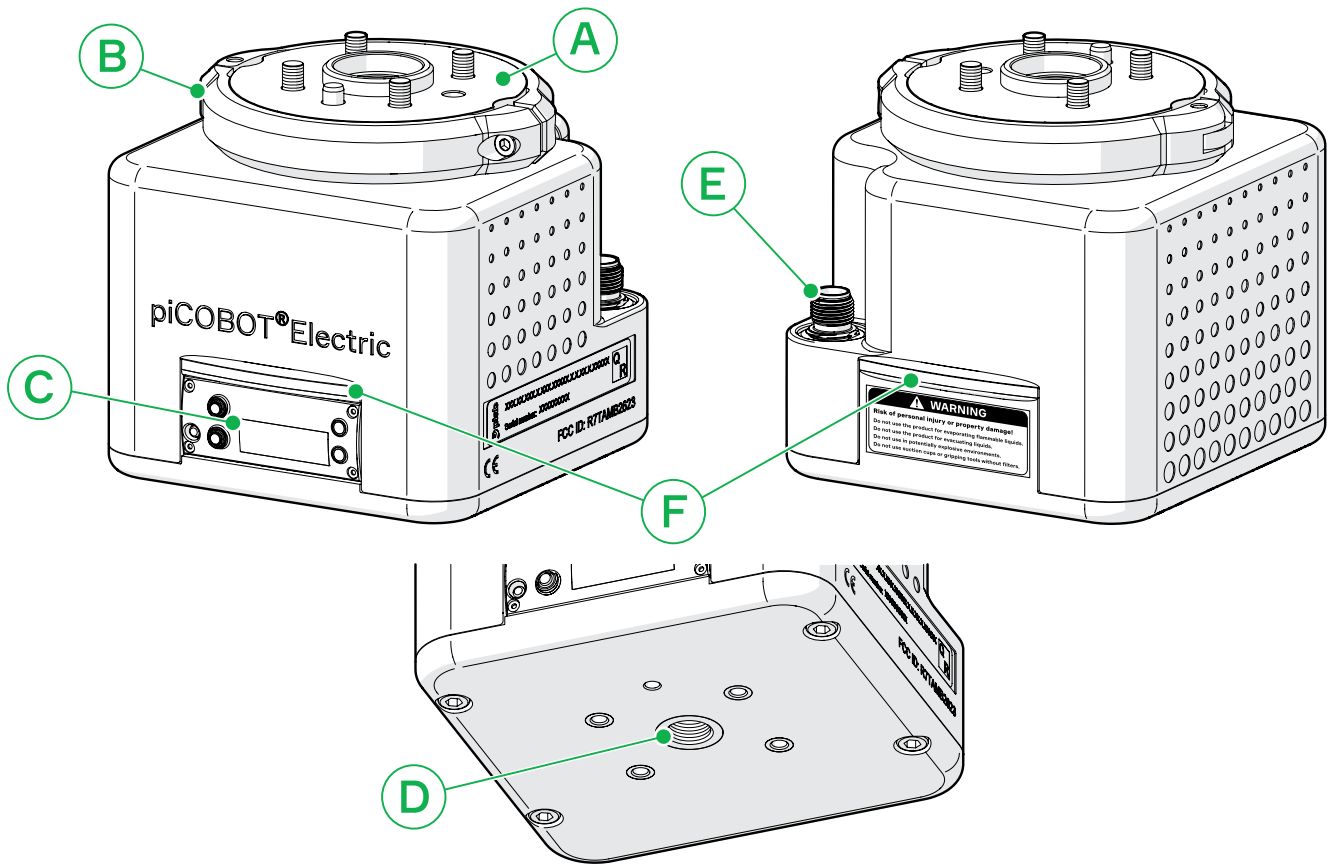


Figure 1. External overview of piCOBOT® Electric.

Position	Description	Note
A	Adapter plate	Adapter plate for robot
B	Clamp ring	-
C	OLED display and select buttons	See Interface section
D	Vacuum port with mesh filter	G1/4" Female
E	Electrical connection	Connector M12 8-pin Male
F	LED light strips	Lights indicate different operation statuses.

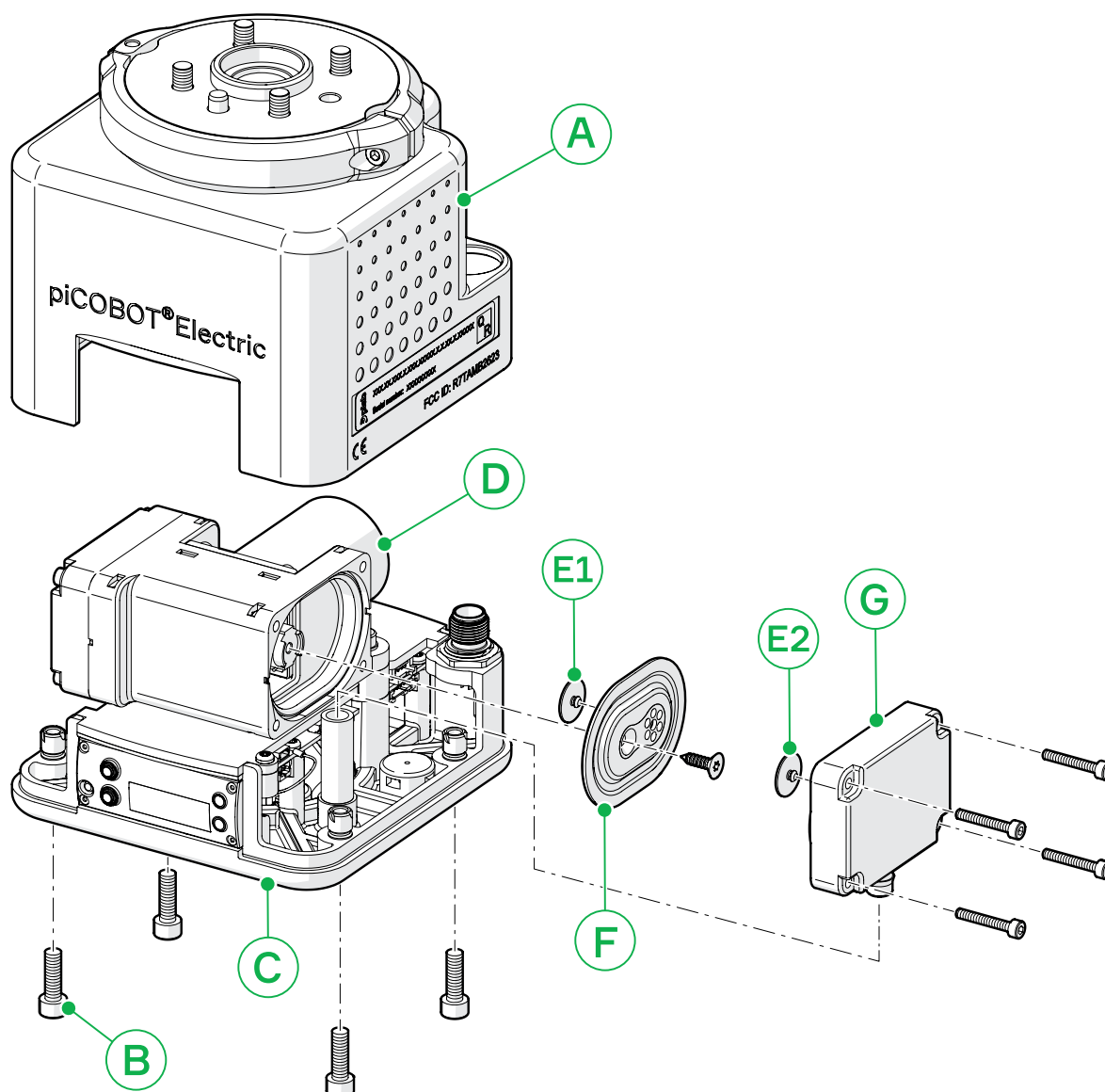


Figure 2. Internal overview of piCOBOT® Electric.

Position	Description	Note
A	Cover	-
B	Cover screws	4 x M5 Hex 4 mm
C	Baseplate	-
D	Piab membrane pump	Vacuum generator
E1 and E2	Non-return valves	The pump has 4 non-return valves in total.
F	Membrane	The pump has 2 membranes, only 1 is shown in the internal overview figure above.
G	Membrane lid	-

4. Installation



Warning

- Do not install or operate your piCOBOT® Electric if damaged during transport, handling, or use. A damaged product may result in bursting and cause injury or property damage.
- Before using the product, read the safety instructions to ensure a safe product operation.

4.1. What's in the box

- piCOBOT® Electric FANUC CRX
- Torx keys (T15)
- Hex keys (2.5, 3, 4 mm)
- Cable M12 8-pin Female Straight to M8 8-pin Female Angled (elbow)
- USB flash drive with FANUC CRX Plugin software
- piCOBOT® Electric FANUC CRX manual
- FANUC CRX Plugin manual
- Adjustable Gripper (optional)
- Suction cups x 4 or x 8 (optional)
- Foam gripper (optional)
- Tool changer (optional)

4.2. Mechanical installation

1. Unpack the box.
2. Use a Hex key (3 mm) to open the clamp ring and release the adapter plate from the pump.
3. Attach the adapter plate to the robot with four MC6S M5 or M6 screws (depending on the selected interface).



Tip

Turn the robot arm so that it is facing upwards, to facilitate the attachment of the adapter plate and reduce the risk of dropping the adapter plate or screws.

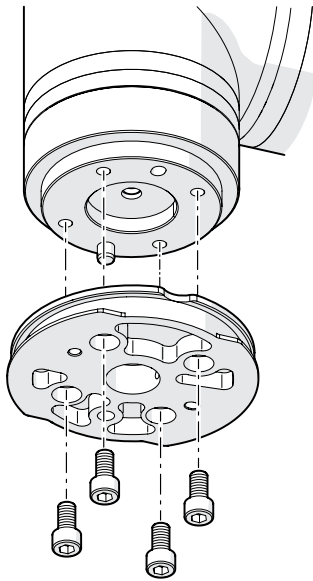


Figure 3. Adapter plate is attached to a robot with 4 screws and a guiding pin.

4. Fasten the piCOBOT® Electric to the adapter plate with the clamp ring. Lock the closed clamp ring with an M4 screw using a Hex key (3 mm). Do not overshoot the maximum torque of 0.7 Nm.

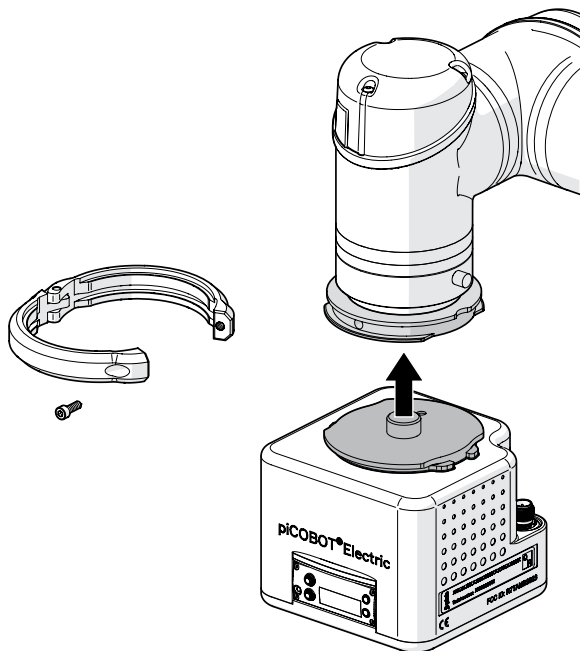


Figure 4. Attach the piCOBOT® Electric to the robot with the clamp ring.

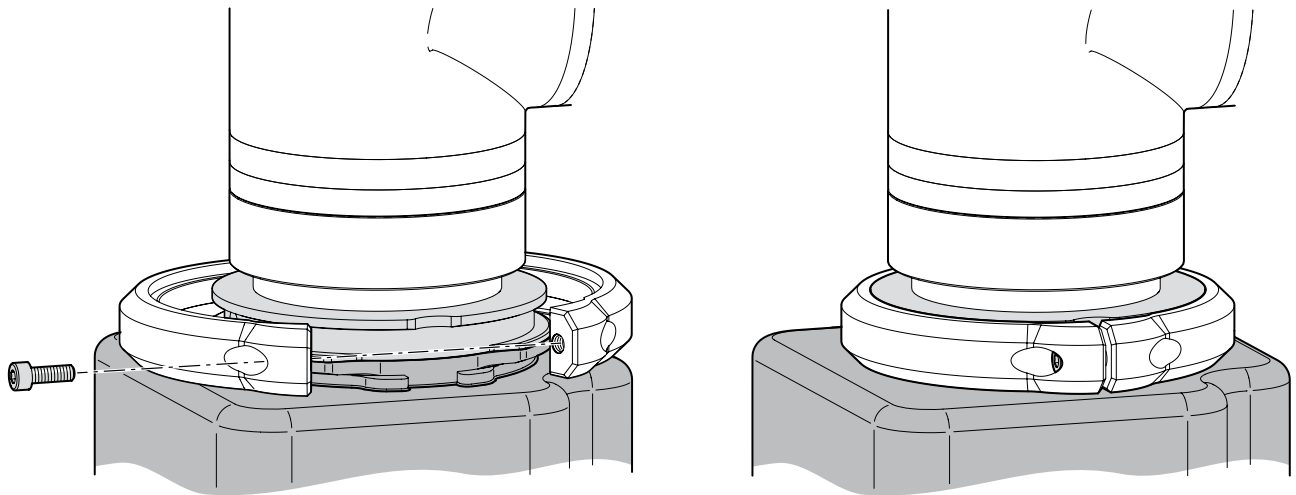


Figure 5. Lock the clamp ring.

5. Mount a gripping tool, such as a suction cup, foam gripper or adjustable gripper to the piCOBOT® Electric, either directly on the tool side or via the tool changer mounted between the piCOBOT® Electric and the gripping tool.

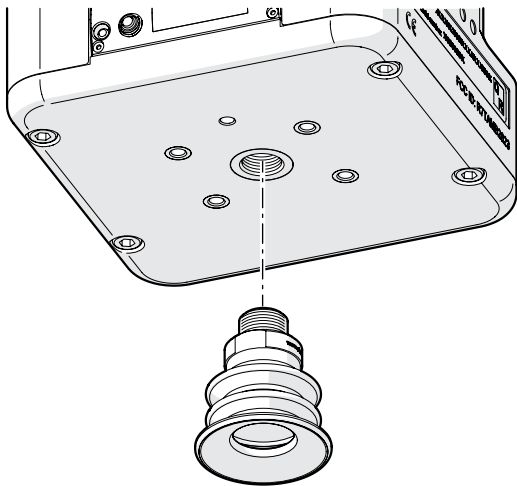


Figure 6. Suction cup mounted on piCOBOT® Electric.

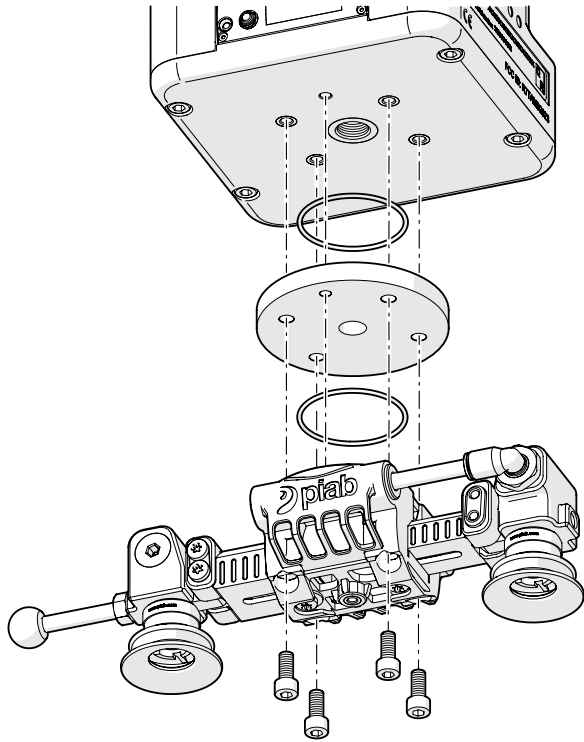


Figure 7. Adjustable gripper attached to the piCOBOT® Electric.

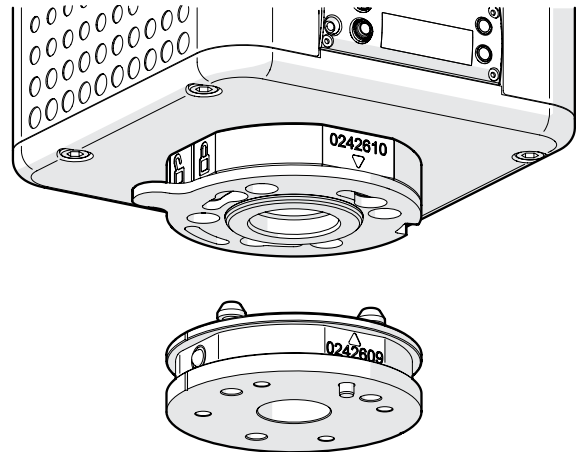
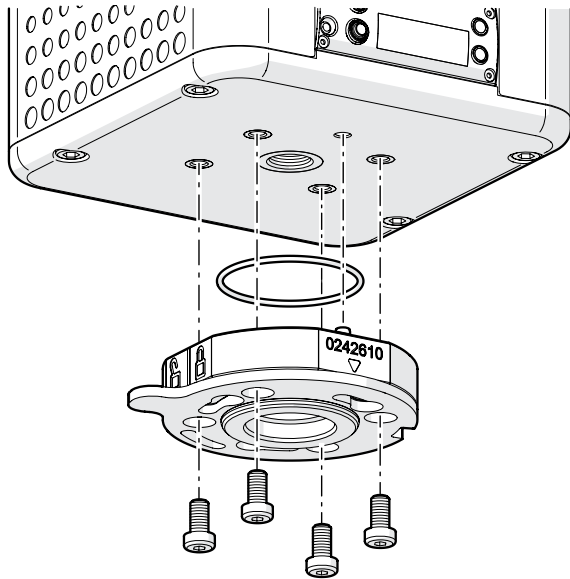


Figure 8. Tool changer attached to the piCOBOT® Electric.

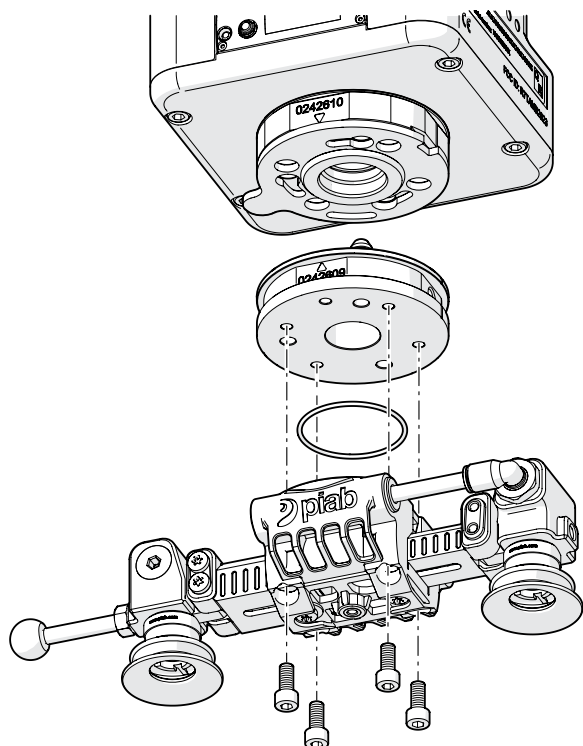


Figure 9. Adjustable gripper attached to the piCOBOT® Electric via the tool changer.

The Tool side of the piCOBOT® Electric has two different mounting interfaces (A and B), this enables fastening different grippers or suction cups:

A - G1/4" Female

B - Interface 30x40 mm, 4 x M5. Recommended torque 1.6 Nm.

For more information see chapter Dimensions.

4.3. Electrical Installation



Warning

Turn off the electrical power to the robot before connecting the electrical cable to piCOBOT® Electric.



Warning

Ensure to use an external slow fuse limited to 3 A for the power supply.

Mount the electrical cable on the piCOBOT® Electric and on the robot, note the guide notches for the right direction.

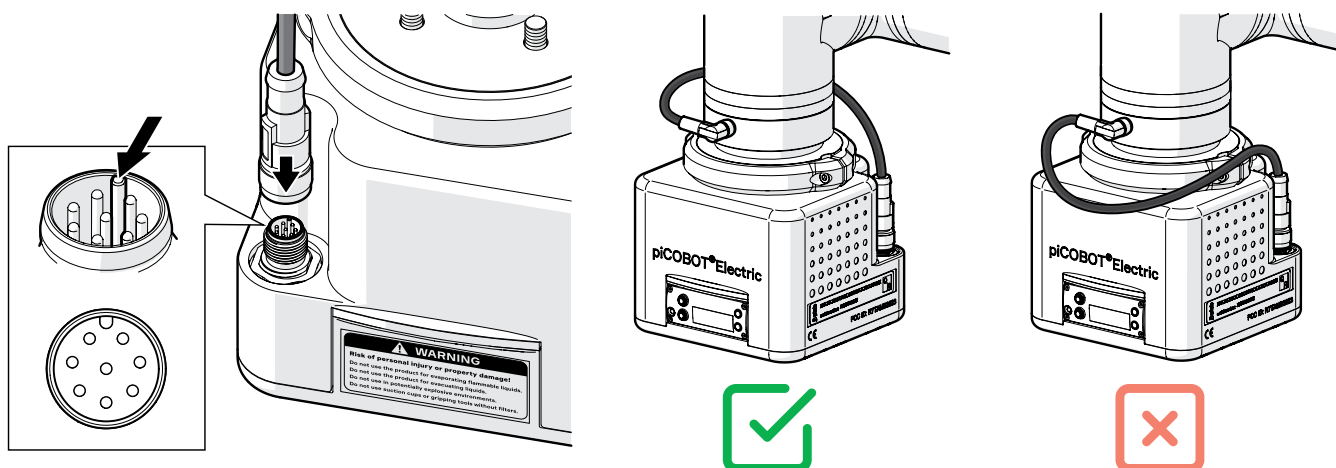


Figure 10. Guide notch on the electrical connector. Correct and incorrect cable arrangement.

4.4. Pneumatic installation



Warning

Use filters in dusty environments

When working in environments with dust, dirt and larger particles that could jam the pump and cause impaired performance or reduced lifetime, use piCOBOT® Electric with gripping tools, such as suction cups or grippers, that are equipped with filters.

4.4.1. Pneumatic diagram

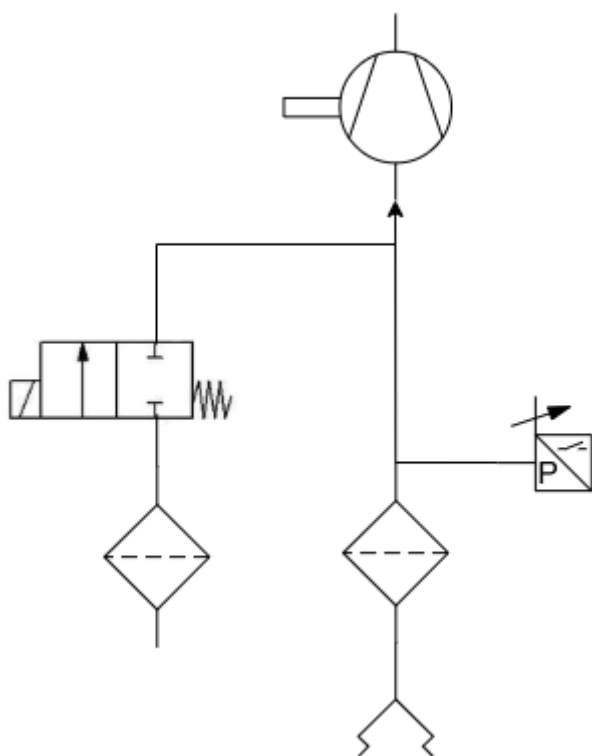


Figure 11. Vacuum NC and Release NC, with non-return valve.

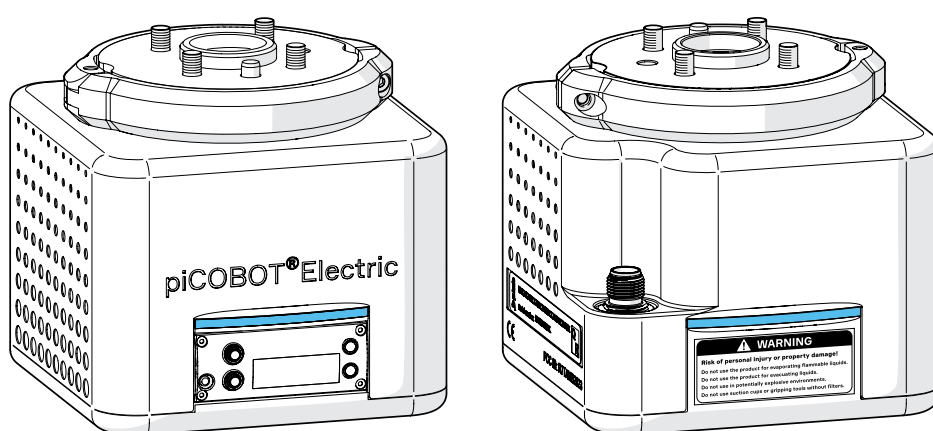
4.5. Software Installation

See the separate FANUC CRX Plugin manual for how to install and use Piab's software plugin for FANUC CRX robots.

5. Operation

The LED will light up in different colors and with steady on or flashing light depending on which task is performed. The two LEDs are located on opposite sides of piCOBOT® Electric.

Task/Activity	Color
Vacuum ON	Blue, flashing
ES Active	Blue, steady
Idle	Green, steady
Release (Release active)	Yellow, flashing



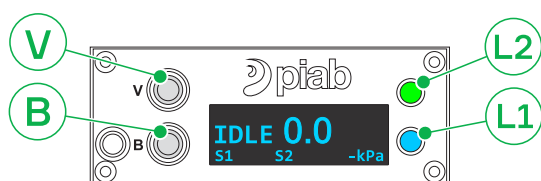
5.1. Operation with FANUC CRX plug-in

To operate the pump, see the separate FANUC CRX plug-in manual.

5.2. Interface

There are two buttons, one OLED screen and two LEDs in green and blue on the piCOBOT® Electric. As soon as the piCOBOT® Electric is turned on, the Piab logo is shown and shortly after the default screen (see picture below, showing the status IDLE with 0.0 -kPa vacuum). The real time vacuum level reading is displayed here when vacuum is achieved. The selected and active vacuum unit is shown in the bottom far-right corner. A screen saver is activated after 3 minutes if no signal in is present or when working in the menu settings.

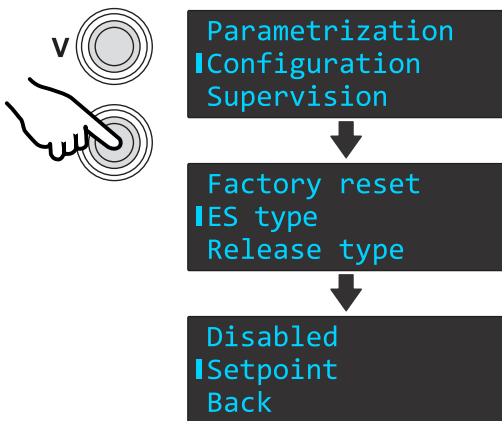
The blue LED L1 will light and IDLE will switch to VAC in the display when an incoming signal for *Vacuum on* is achieved. Likewise, for a *Release on* signal but then the green LED L2 is lit and IDLE switches to *Release* on the display. It is also possible to manually operate *Vacuum on* or *Release on* with the buttons “V” (*Vacuum on*) and “B” (*Release on*) when the piCOBOT® Electric is powered with 24V. *Release on* will always override *Vacuum on*.



The information in the bottom of the display shows, from left to right, the setpoint status Part Present (PP); Energy Saving (ES) level achieved; and the vacuum unit. The top left section in the display will show status conditions, e.g. when and which Energy Saving function or which Release type is active and in action. The output signals and status conditions can represent different selectable features, for more info see the Menu overview and Menu setting on the coming pages.

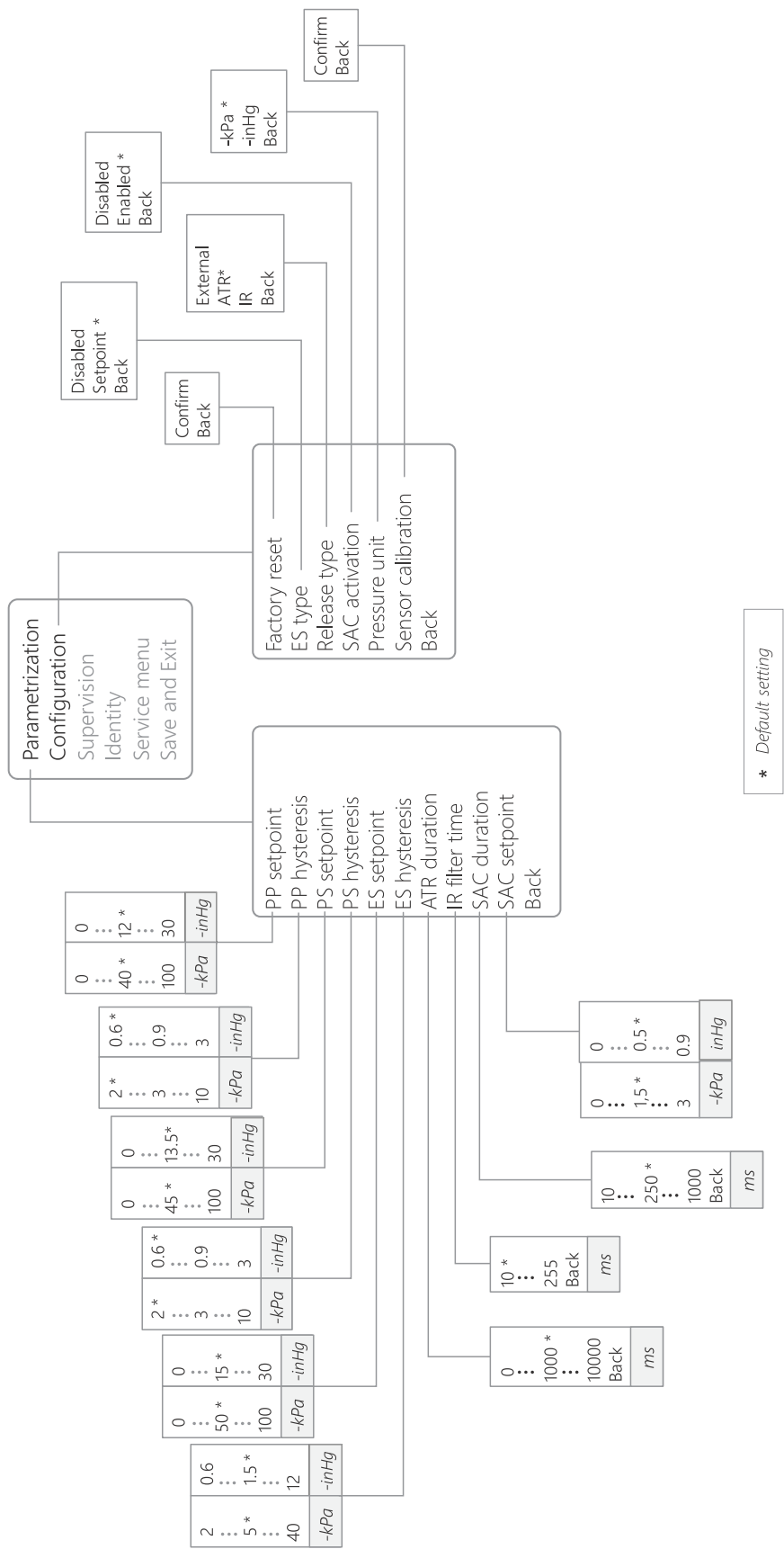


To enter the menus, press and hold “V” and “B” at the same time for 2 seconds. A small time-bar on the top of the display will gradually go from left to right until menu appear.



To navigate the menus, push “B” to toggle down and “V” to toggle up the list. Hold to scroll the list. Set your choice to the square marker and press both “V” and “B” at the same time. To select and set a value or choice in a menu, toggle to the square marker and press “V” and “B” at the same time again. Your choice is marked with * and the OLED will quickly blink to confirm your setting. In the Parametrization menu wait for 2 seconds to leave the entered menu. All other menus have a “Back” function at the bottom of different selections. After 10 seconds without new input the screen automatically returns to its default position.

5.3. Menu overview piCOBOT® Electric FANUC CRX



5.4. Menu settings

Parametrization

Parametrization	Menu name	Preset value	Range	Unit	Comment
Part Present setpoint	PP setpoint	40 [12]	0...100 [0...30]	-kPa [-inHg]	Normally used as Part present level.
Part Present hysteresis	PP hysteresis	2 [0.6]	2...10 [0.6...3]	-kPa [-inHg]	-
Part Secured setpoint	PS setpoint	45 [13.5]	0...100 [0...30]	-kPa [-inHg]	-
Part Secured hysteresis	PS hysteresis	2 [0.6]	2...10 [0.6...3]	-kPa [-inHg]	-
Energy Saving setpoint	ES setpoint	50 [15]	0...100 [0...30]	-kPa [-inHg]	Normally used as Energy Saving trigger level.
Energy Saving hysteresis	ES hysteresis	5 [1.5]	2...40 [0.6...12]	-kPa [-inHg]	-
Automatic Timer Release duration	ATR duration	1000	0...10000	ms	-
Intelligent Release filter time	IR filter time	10	10...255	ms	-
Self Adhesion Control duration	SAC duration	250	10...1000	ms	Time for one release actuation (in milliseconds).
Self Adhesion Control setpoint	SAC setpoint	1.5 [0.5]	0...3 [0...0.9]	-kPa [-inHg]	-

BACK - Back exits the parametrization menu and returns to the top menu.

Configuration

Function	Menu name	Preset value	Configuration	Comment
Factory reset	Factory reset	-	Confirm	See corresponding chapter Function.
Energy Saving type	ES type	Setpoint	Disabled Setpoint	See corresponding chapter Function.
Release type	Release type	ATR	External ATR IR	See corresponding chapter Function.
Self Adhesion Control activation	SAC activation	Enabled	Disabled Enabled	See corresponding chapter Function.
Pressure unit	Pressure unit	-kPa	-kPa -inHg	See corresponding chapter Function.
Sensor calibration	Sensor calibration	-	Confirm	Resets and calibrates the analog sensor.

BACK - Back exits the configuration menu and returns to the top menu.

Supervision

Function	Menu name	Unit	Comment
Vacuum cycle counter	Vac cycle counter	-	See corresponding chapter Function.
System Temperature	Sys temp	°C	See corresponding chapter Function.
Time to Part Present	Time to PP	s	See corresponding chapter Function.
Average Time to Part Present	Av. Time to PP	s	See corresponding chapter Function.
Time to Part Secured	Time to PS	s	See corresponding chapter Function.
Average Time to Part Secured	Av. Time to PS	s	See corresponding chapter Function.
Time to Energy Saving	Time to ES	s	See corresponding chapter Function.
Average Time to Energy Saving	Av. Time to ES	s	See corresponding chapter Function.
Time running (in minutes)	Time running	min	See corresponding chapter Function.
Average motor speed	Av. RPM	rpm	See corresponding chapter Function.

BACK - Back exits the supervision menu and returns to the top menu.

Identity

Function	Menu name	Example value
Device ID	Device ID	321
Product code	Product code	PCOE.000000000000
Manufacturing date	Manufacturing date	2024-05-01
Serial number	Ser No.	24Q001234
Vendor	Vendor	Piab
Hardware revision	HW rev.	
Firmware revision	FW rev.	

BACK - Back exits the identity menu and returns to the top menu.

Service menu

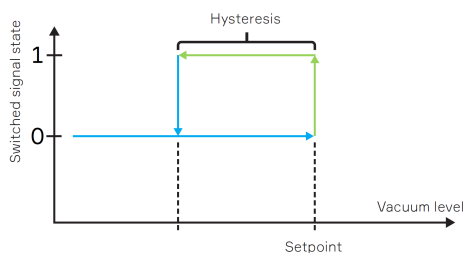
Function	Menu name	Configuration
Hours to membrane service	hrs to service	2135 (example data)
Reset membrane service countdown	Service reset	Confirm
Display membrane service countdown	Show countdown	Disabled Enabled
Piab lock (for Piab developers only)	Piab lock	Internal only

BACK - Back exits the service menu and returns to the top menu.

6. Functions

6.1. Switched vacuum levels: PP, PS and ES

Switched signals are standard “digital sensor feedback signals” using a target value and a hysteresis to be defined. The signals are transmitted to the control system through the RS-485 process data. The operation of the switched signals is according to the figure below.



6.1.1. Part Present (PP)

Part Present (PP) is used as an indicator that the part was successfully picked. PP is set to value low when vacuum level drops below the value set for Part Present hysteresis.

6.1.1.1. Time to Part Present

Time to Part Present is the time from when vacuum is turned on until the system reaches the configured Part Present vacuum level.

6.1.1.2. Average Time to Part Present

Average Time to Part Present is the rolling average value of the last 20 samples of Time to PP. If values within the last 20 picks have been invalid (value=0), the samples will be omitted from the average value.

6.1.2. Part Secured (PS)

Part Secured (PS) can be used as a confirmation of that the part is secured and the robot or machine is ready for motion at full speed. PS is set to low when the vacuum level drops below the value set for Part Secured hysteresis.

6.1.2.1. Time to Part Secured

Part Secured is usually the middle signal level of the Setpoints, used for the end-user to identify when a motion can be initiated by robot or machine.

6.1.2.2. Average Time to Part Secured

A rolling time average parameter is available for the Part Secured setpoint.

6.1.3. Energy Saving (ES)

Energy Saving (ES) functions reduce energy consumption and noise levels whenever the application and conditions allow for it. User can set the target vacuum level and the product adjusts the flow capacity and minimizes the current consumption needed to reach and maintain that target vacuum level.

6.1.3.1. Energy Saving setpoint

Energy Saving Setpoint is a pre-set threshold level for Energy Saving. ES setpoint is set to low when vacuum level drops below the ES setpoint hysteresis.

Design recommendation - Running with an unnecessarily deep ES setpoint level will force the ejector to operate much longer during the evacuation cycle than needed. Therefore, the setting of ES shall be done with the required vacuum level margins for PP and PS in mind.

6.1.3.2. Time to Energy Saving

A timer is connected to the Energy Saving signal.

6.1.3.3. Average Time to Energy Saving

A rolling time average parameter is available for the Energy Saving setpoint.

6.2. Release

Release functions are used to release a handled object quickly by opening the valve to atmospheric pressure.

Release (always pneumatically normally closed) is activated by an “active” input signal. Active Release control will always override Vacuum control, terminating the vacuum generation and starting the release. For related information, see Vacuum control on previous page.

6.2.1. External control

External control is primarily selected and used when the user wants complete control of the release from the main process controller (PLC) or robot controller and the timing of that controller is considered sufficient.

6.2.2. Automatic Timer Release (ATR)

Automatic Timer Release (ATR) means that atmospheric release function will start automatically after the vacuum control signal is set to "vacuum OFF" by the external control (i. e. vacuum state shifts to "vacuum OFF"). The duration is set with a timer, range 0-10 000 milliseconds.

Use case – ATR makes programming easier by eliminating the need to add the extra step of sending a release command for a specified duration.

6.2.3. Intelligent Release (IR)

Intelligent Release (IR) is an alternative to reduce the need for tuning of part release. IR will start automatically when the vacuum control signal is set to "vacuum OFF" by the external control. The release duration is optimized, and will automatically stop when all vacuum is removed from the system. IR is a self-learning function and only needs a few cycles to optimize release duration for different system volumes. In the initial cycles, an extra release-activation can be presented to fully remove vacuum. IR filter time is used to tune the release depending on the size of the vacuum system. piCOBOT® Electric will send a signal (APA) when atmospheric pressure is achieved.

IR filter time: range 10-255 ms. Low number = the most energy-efficient release (suitable for small vacuum system). The higher the number the larger or more flow-restricted the system.

Use case – IR makes programming easier as by eliminating the need to add the extra step of sending a release command with a user-specified duration. Since the IR will adapt to fluctuations or variations in the application, it will always ensure a successful release. All that is required is that you move the handled object to the target drop-off point and await the “Release complete” signal before transferring back for the next pick. Using a standard external release control, the user would typically set a longer release time than necessary in order to secure a stable release and return motion of the robot or machine. IR allows the user to run just the right amount of release needed for the specific cycle without the need for detailed fine-tuning and adjustment.

6.3. Self Adhesion Control (SAC)

Self Adhesion Control (SAC) automatically removes “unwanted” vacuum with short actuations of the release valve if the vacuum ON signal has not been activated.

SAC duration is the time for one release actuation (in millisecond). SAC setpoint: Low value = sensitive system for very light handled object, but at the risk of "ghost" actuations due to signal drift/disturbance. High value = robust but not as responsive system.

Use case - Unwanted vacuum is typically created when a compressed suction cup is expanded by moving the gripper tool away from the handled object with vacuum turned off (during the placing of the object). SAC is activated if such unwanted vacuum is created, in order to break vacuum and avoid dragging the object with the motion of the gripper tool.

6.4. Supervision and Diagnostics Functionalities

6.4.1. Vacuum level

The vacuum level is measured by the pump. The sensor covers a range of -100 to 100 kPa. Vacuum values are always reported as "positive" values since the unit of measurement is [-kPa, -inHg]. Positive pressures (overpressure) are subsequently reported as negative values.

6.4.2. Sensor calibration

Zero-point calibration of sensor procedure (pressure range 1 atm $\pm$ 10 kPa):

1. Make sure the pump is in an IDLE state.
2. Navigate on the display menu to Sensor calibration, located under Configuration.
3. Press Confirm. The sensor is now calibrated to zero.

Sensor data inaccuracies can usually be avoided with proper sensor calibration and using additional filtration in dusty environments.

6.4.3. Pressure unit

The pressure unit can be toggled between -kPa and -inHg.

6.4.4. Vacuum cycle counter

The vacuum cycle counter counts the number of instances that vacuum has been turned OFF throughout the lifetime of the pump.

6.4.5. System Voltage

System Voltage is the latest real-time reading of voltage.

6.4.6. System Temperature

This is the latest reading of system temperature. The system temperature is the circuit board temperature. Temperatures above 100 °C indicate that there is a problem either with the installation or with the pump.

6.4.7. Service menu

6.4.7.1. Time running

Time (in minutes) that the electrical motor has been running.

6.4.7.2. Average RPM

Average Revolutions Per Minute (RPM) of the electrical motor during its vacuum ON cycles calculated throughout the lifetime of the pump.

6.4.7.3. Hours to membrane service

The device computes and displays the number of hours left before the membranes of the electrical pump need service (replace with new membranes). The device issues a warning on the display when 100 hours of operation remain before the membrane service. The warning does not interfere or influence the current tasks of the device. It is possible to disable the warning in the service menu of the device.

6.4.7.4. Reset membrane service countdown

After replacement of membranes, the membrane service countdown timer should be reset to zero.

6.4.7.5. Show membrane service countdown

The device display shows a warning and a countdown starting from when 100 hours remains to membrane service.

6.4.7.6. Factory reset

Factory reset returns the product to the values set for parameterization and configuration parameters based on the product code.

7. Maintenance



Warning

Prior to any maintenance work, turn off or disconnect the electrical power supply and ensure that the motor of the pump has cooled down.

The piCOBOT® Electric has been designed to minimize the need of maintenance. To ensure safe operation the following actions are strongly recommended to take:

- Maintenance must be performed with a fixed period short enough, to ensure a firm grip at all times.
- An overall inspection of the piCOBOT® Electric must be performed on a regular basis with a duration depending on the application characteristics.
- All maintenance must be performed in compliance with this manual, including safety instructions.
- Only authorized integrators, or Piab AB, shall perform repairs.
- Only use original spare parts.

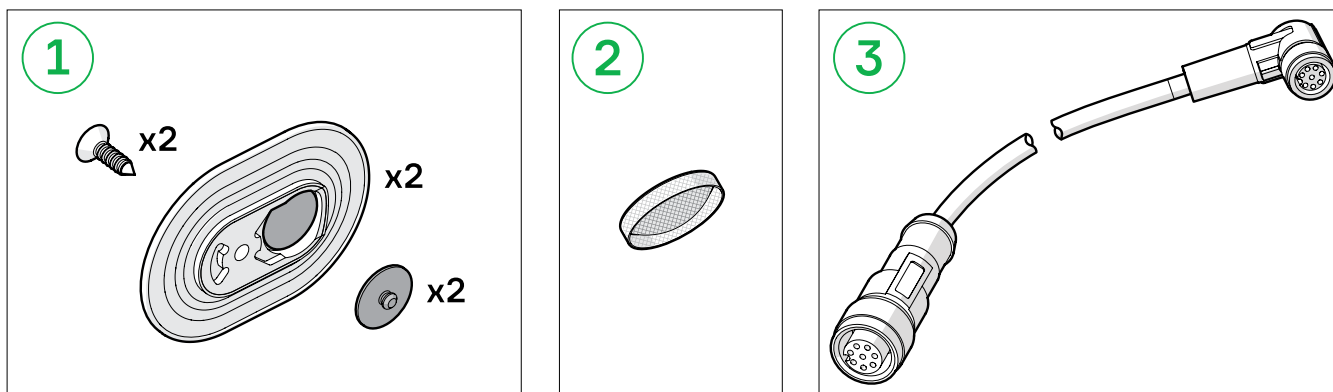
7.1. Preventive maintenance

The following preventive maintenance actions are recommended to take:

Parts for maintenance	Item no	Action	Service action and/or inspection interval	Further action if unresolved
Adjustable gripper	0244112	Inspect	After every 200 hours of operation.	Replace gripper if locking of arms is not possible.
Foam gripper	See piab.com	Clean	Clean micro holes every month.	Replace with new foam.
Suction cups	-	Inspect	When/if poor gripping performance.	Replace suction cup.
Electrical cable	0246697	Inspect	After 500 installations/removals.	Replace cable.
Electrical vacuum pump membranes	0247361	Replace	Replace membranes when pump indicates time for membrane service (see chapter 6.5). Reset the membrane counter after the membrane change. The frequency of replacement depends on the application and operation of the pump.	-

** Free running vacuum is the vacuum level displayed when the suction cups are not in contact with the work object and piCOBOT® Electric is generating vacuum.

7.2. Spare parts



Position	Item number	Quantity	Description
1	0247361	1	piCOBOT® Electric Service Kit: Membranes (x2), non-return valves (x4), screws 3.5x12 mm (x2 pieces)
2	0123110	1	Mesh filter
3	0246697	1	Cable M12 8-pin Female Straight to M8 8-pin Female Angled (elbow)

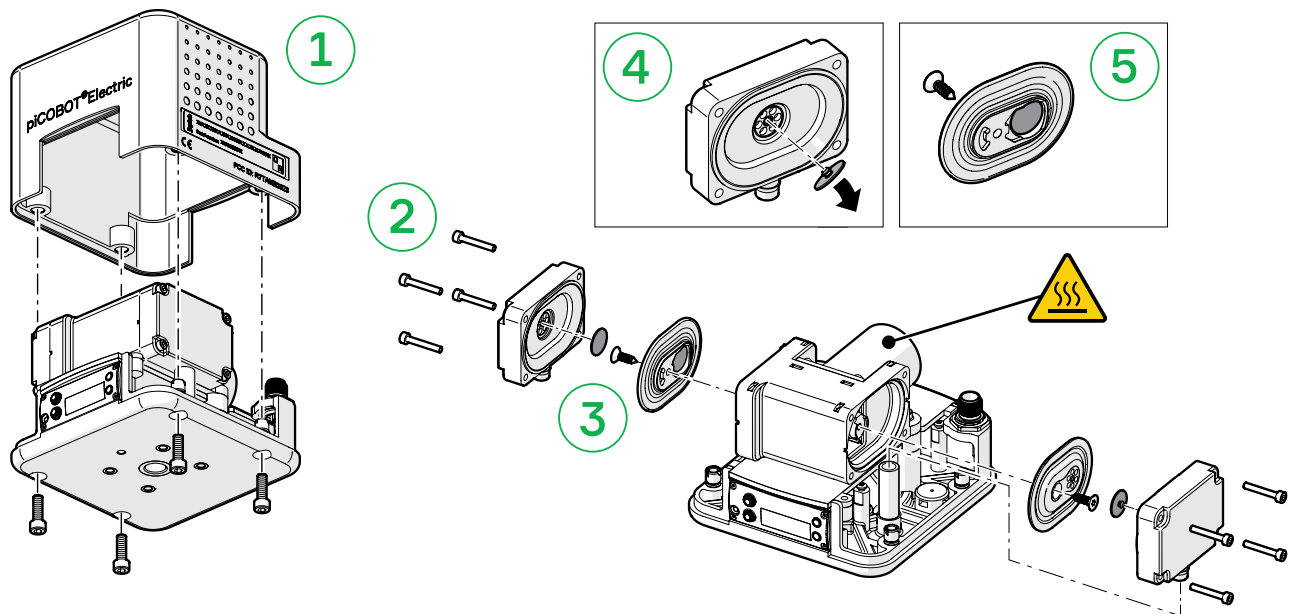
7.3. Replacing the membranes and non-return valves



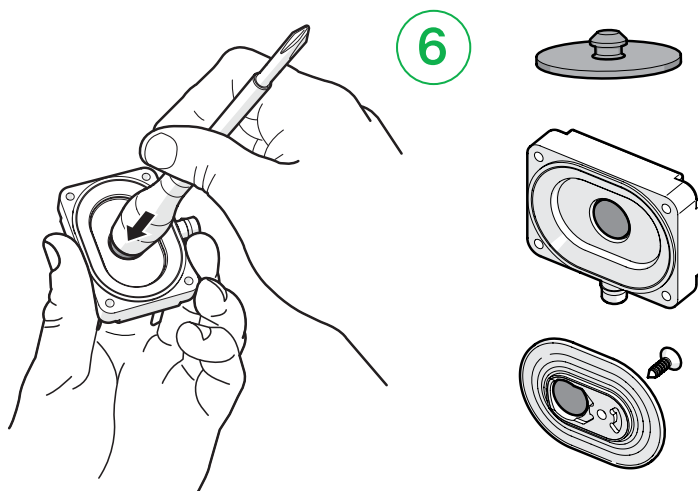
Warning

Hot motor surface can cause severe injuries, ensure the pump has cooled down prior to disassembly.

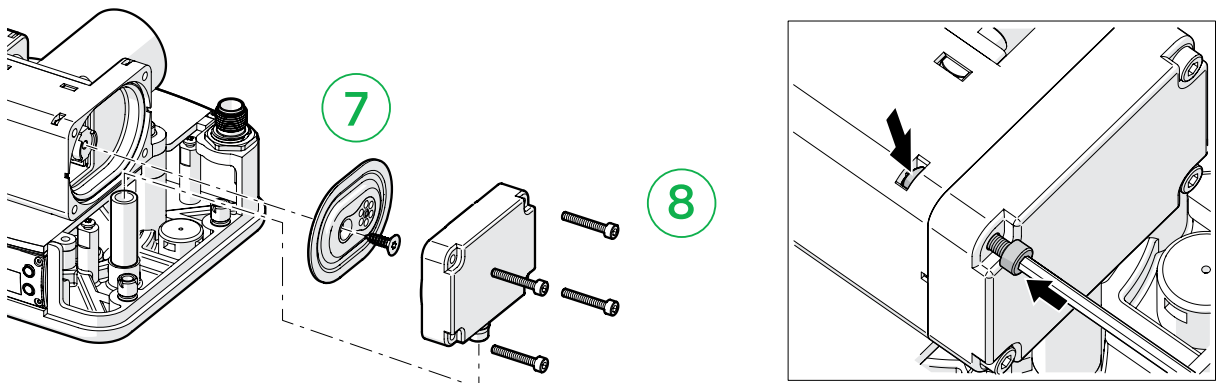
1. Remove the safety cover for the piCOBOT® Electric house, unscrew the four Hex 4 mm M5 screws holding the cover and baseplate together.
2. Remove the membrane lid, unscrew the four Hex 2.5 mm M3 screws, and disconnect the membrane lid from the silicone tube. Let the silicone tube remain connected to baseplate.
3. Remove the membrane by unscrewing the T15 3.5 mm screw.
4. The outer non-return valve is attached to the center hole of the membrane lid. Remove the non-return valve by hand. Replace it with a new one.
5. The membrane has a non-return valve attached to the center hole. Replace the membrane and non-return valve with new parts.



6. Use the top of the handle of a screwdriver to push the non-return valve back into the center hole of the membrane lid.



7. Attach the membrane with the T15 screw, recommended torque: 0.6 Nm. Make sure that the screw is inserted into the hole of the pump piston.
8. Connect the membrane lid to the silicone tube and attach the membrane lid to the pump with the four screws. Each screw should fit into a square nut, if not adjust the nut position. Recommended torque: 1.0 Nm.

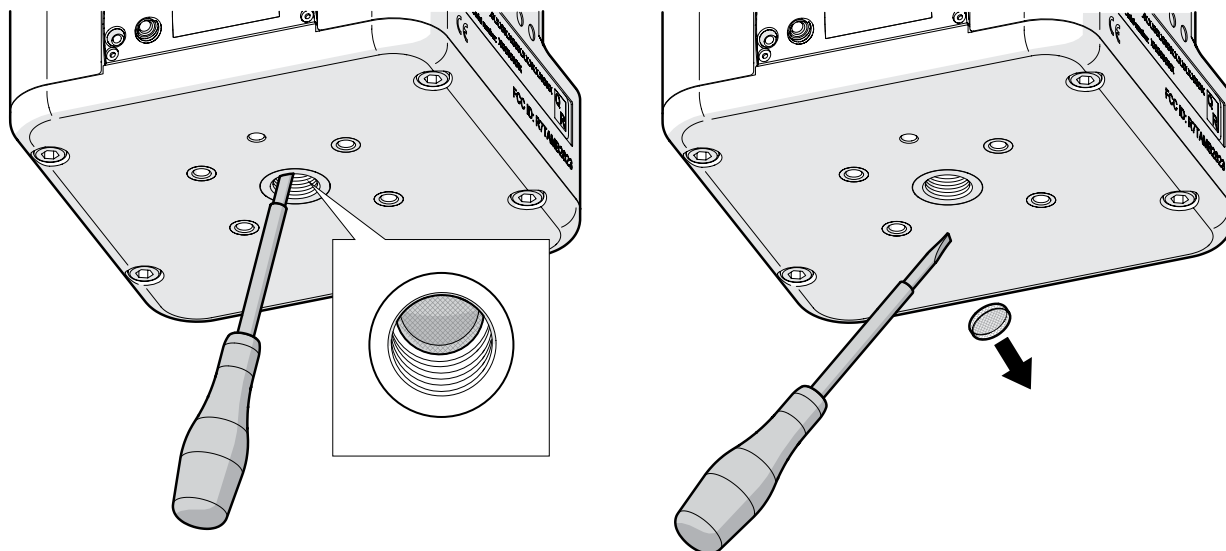


9. Attach the safety cover to the baseplate with four M5 screws, the recommended torque is 1.6 Nm.

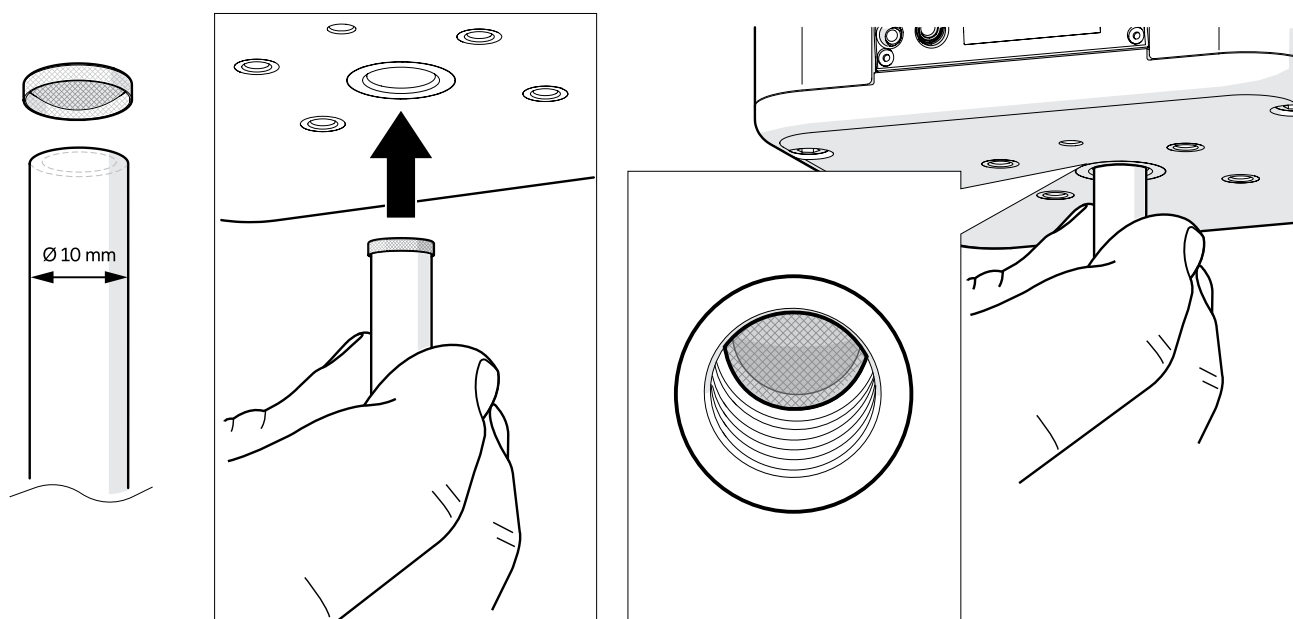
7.4. Replacing the vacuum filter

The base plate contains a mesh filter, that can be replaced.

Use a screwdriver or equivalent to pull the mesh filter out from the vacuum port.



Use the end of a 10 mm hose to push the new vacuum filter inside the base plate of piCOBOT® Electric.



7.5. Troubleshooting

Observation or error	Subsystem	Actions	Comment
No vacuum in system	Pump	Inspect power supply	Power supply is monitored on the display and shall be within $24\text{ V} \pm 10\%$
Poor vacuum in system	Pump	Check for leakage	Unexpected leakage in the system may cause the pump to make unexpected noise or cause excess heat.
	Suction cup	Check condition of suction cups	Look for cracks and wear. Correct suction cup for the object?
	Suction cup	Clean filter in suction cups	Clogged filter can manipulate "ready to go signal" in pump
	Suction cup General	Clean piSAVE® sense valve Check O-ring sealing between pump and gripper	Enables "double pick" feature Look for cracks and wear. Use correct torque for connection screws (4 screws)
Poor gripping	Pump	Adjust ES Setpoint levels	Too low vacuum level gives less lifting force.
	Suction cup	Check angle and adjustment on suction cups	-
	Suction cup	Revise the choice of suction cup	The gripping application might need other type of suction cups
	Foam gripper	Clean micro holes	-
	Foam gripper	Clean foam filter (for filter version grippers)	-
	General	Lift is too slow. Slow down lift, avoid acceleration peaks.	-
Ghost signal. Part Present comes without object in contact	Suction cup	Clean filter in suction cups	-
	Foam gripper	Clean micro holes	-
Robot goes into safety mode	Pump	Reset robot	Short circuit in the robot due to uncared use will force it into safety mode.
Pump is making noise	Pump	Check for leakage	Unexpected leakage in the system may cause the pump or motor to make unexpected noise.
Abrupt loss of function	Pump	Check connections	Connections not fastened properly during installation may come loose during operation.
Foam wears too quickly	Foam gripper	Reduce wear by gripping parallel to the work piece surface.	-
Parts are not released	Pump	Review release settings and possibly adjust. See ATR and IR settings in chapter 6.1	-
Motor stalled	Motor	Contact Piab's customer support	The motor should not stall and if this error appears, please contact Piab customer support for further actions.

7.6. Accessories

Item no	Description
0244112	Adjustable gripper 30x40
0101431	Suction cup B30-2 Silicone, G1/4" male, with mesh filter
0205130	Suction cup B52XP Polyurethane 30/60, G1/4" male, with mesh filter
0114449	Suction cup BX35P Polyurethane 30/60 with filter, G1/4" male, with mesh filter
9914199	Suction cup piGRIP® with Bag lip Silicone, G1/4" male, with mesh filter
See piab.com	Other suction cups with G1/4" Female fitting are available
9979763	Foam gripper 140x140x10 mm, PCOF.140.140.N114.CV07.DM
9979764	Foam gripper 140x140x10 mm, PCOF.140.140.N114.FR6.DM
0242590	Tool changer complete
0242609	Tool changer - Tool side
0242610	Tool changer - Pump side
0243240	Spare part kit O-ring - Tool changer
0243619	Docking station 105-125
0243620	Docking station 105-165

8. Technical Data

Description	Unit	Value
Installation		
Total weight (with gripping tools)	g [oz]	See the configurable datasheet, available on piab.com, for the weight of the particular configuration.
piCOBOT® Electric weight (including adapter plate)	g [oz]	1100 [38.8]
Maximum payload	kg [lb]	16 kg [35.2 lb]
Material	-	PA, NBR, SS, Al, FPM, CuZn, Cu, PU, Q (Silicone), PMMA
Connection, electrical	-	Connector M12 8-pin Male
Maximum current limit	mA	1100
Peak current	mA	1500
Peak current time	ms	<5
Connection, vacuum	-	G1/4" Female
Environmental properties		
Safety classification	-	IP54
Temperature range	°C [°F]	0-40 [32-104]
Humidity range	%RH	35-85
Vibration resistance at 2g xyz	Hz	8-200
Noise level, pump with suction cup	dBA	65
Noise level, pump with foam gripper	dBA	61
Operation		
Maximum vacuum	-kPa [-inHg]	70 [21]
Maximum vacuum flow	l/s [scfm]	0.79 [1.67]
Release flow (atmospheric release)	l/s [scfm]	0.35 [0.74] at 50 -kPa/14.81 -inHg
Release function operation	-	NC solenoid release valve
Display	-	OLED and gyro display
Service interval	-	Membranes must be changed regularly. The frequency of replacement depends on the application and operation of the pump. For reference, in a normal pick and place application with 50 -kPa set point, around 1 500 000 cycles could be expected.
Electrical input/output		
Supply voltage	VDC	24 ± 10%
Communication interface	-	RS-485
Sensor feedback	-kPa [-inHg]	0-100 [0-29.61]
Sensor accuracy of full scale analog output	-	±3%
Manual override	-	Yes, non-locking push style
Response time valve	ms	20
Switch output S1/S2, max.	mA	2x40 simultaneously, or 1x80 one at a time

8.1. Vacuum performance

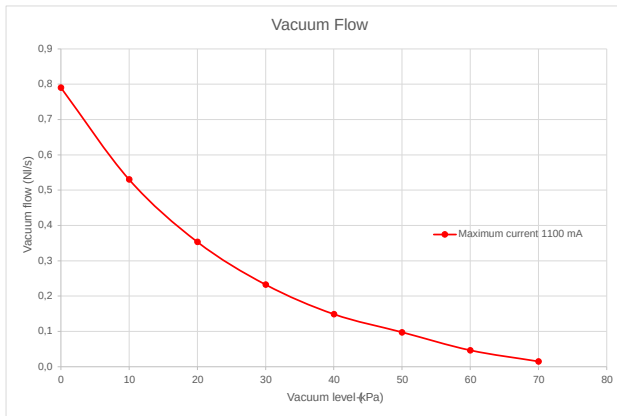


Figure 12. Vacuum flow

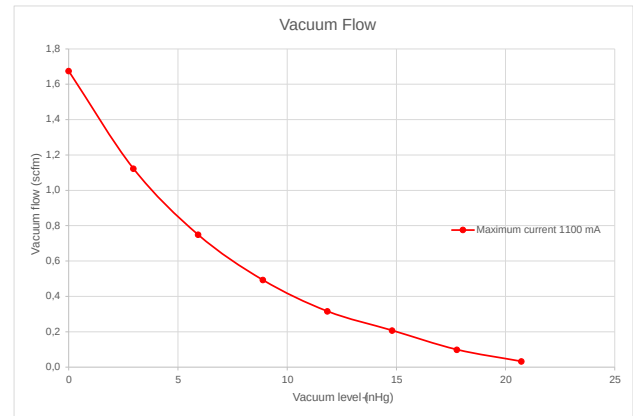


Figure 13. Vacuum flow (Imperial units)

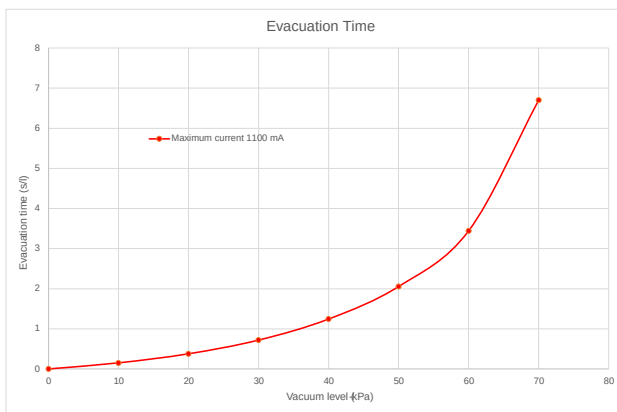


Figure 14. Evacuation time

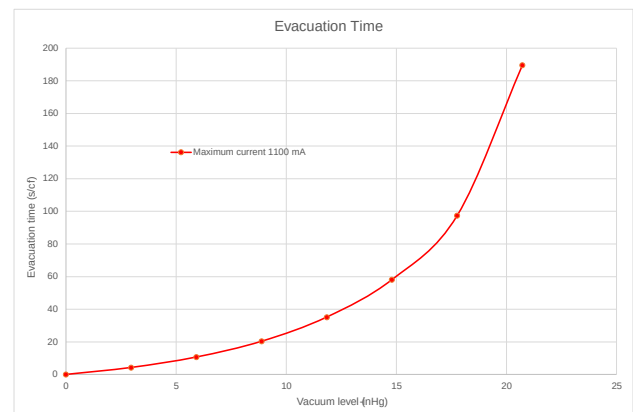


Figure 15. Evacuation time (Imperial units)

8.2. Current consumption

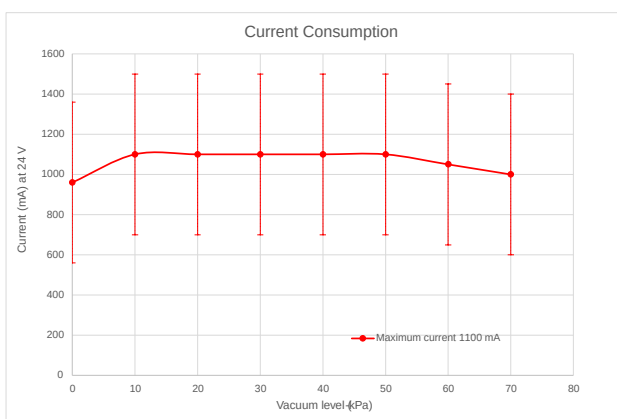


Figure 16. Current consumption

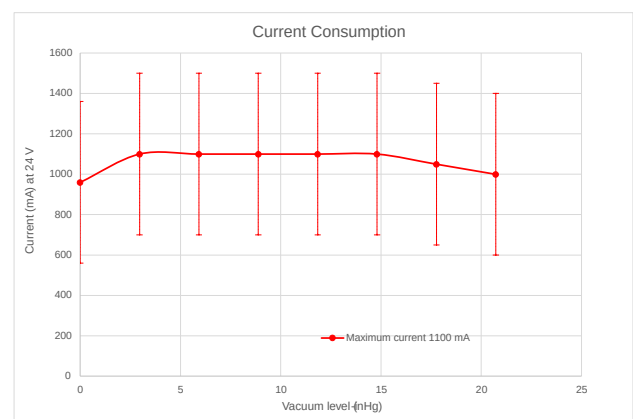


Figure 17. Current consumption (Imperial units)

Current consumption (red lines in the figures above) is the real-time electrical power piCOBOT® Electric uses, while peak current (vertical red dotted lines in the figures above) consumption is the maximum power the piCOBOT® Electric draws at any point. Peak current is crucial for designing electrical systems to handle the highest possible load safely and efficiently.

8.3. Efficiency

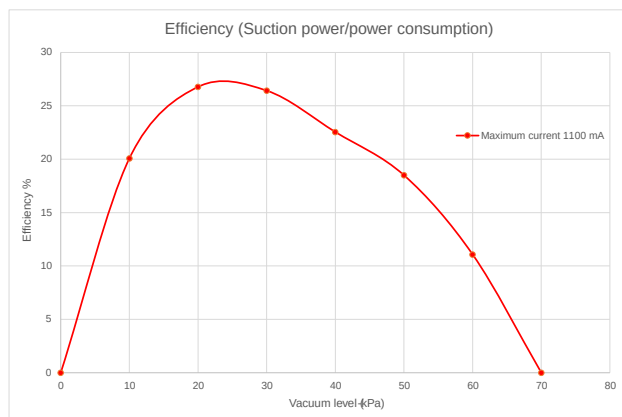


Figure 18. Efficiency of piCOBOT® Electric: suction power versus robot power consumption.

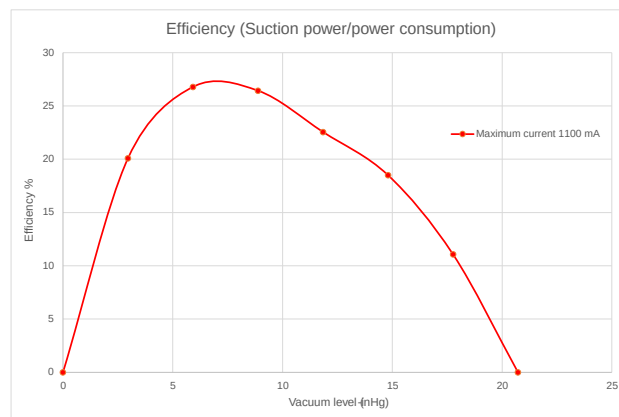
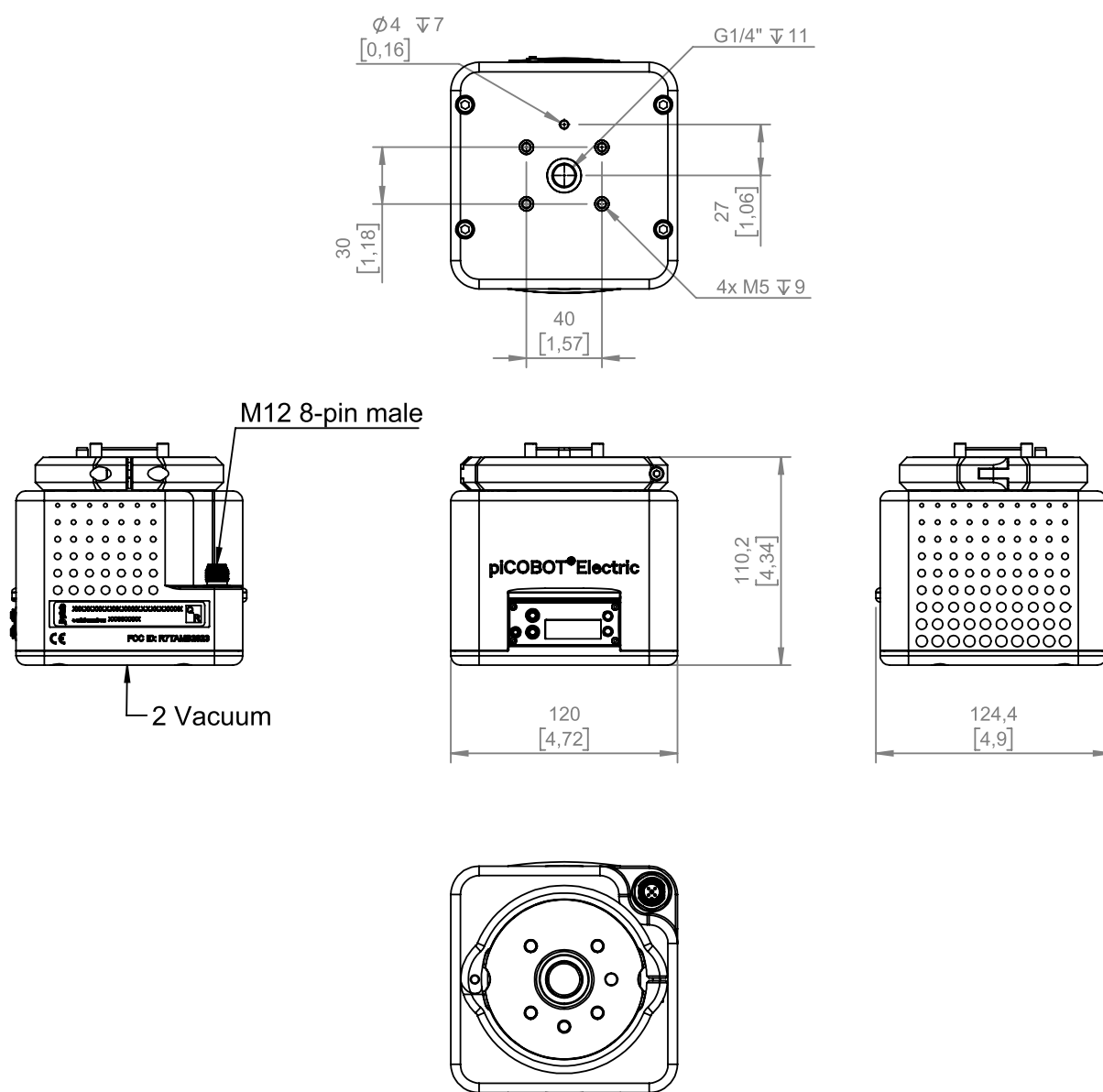


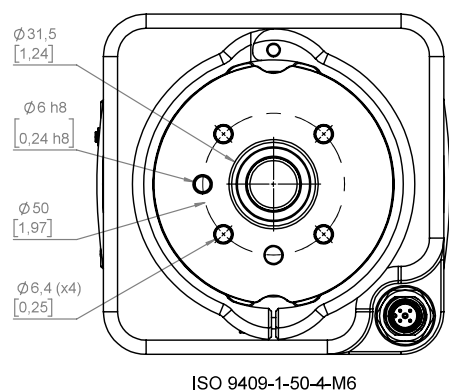
Figure 19. Efficiency (Imperial units)

9. Dimensions



9.1. Adapter plates

Dimensions



10. Warranty

The Seller gives its Customers a five-year warranty from the receipt of the Products for vacuum pump Products (excluding vacuum pumps with electronics/controls, electro-mechanical vacuum pumps, accessories, and controls).

The Seller gives its Customers a one-year warranty from the receipt of the Products for all other Products (i.e. excluding vacuum pump Products but including vacuum pumps with electronics/controls, electro-mechanical vacuum pumps, accessories, and controls) if the failure has occurred within the specified lifetime in terms of duty cycles as set out in the Product specification (if any).

The warranty covers manufacture and materials defects in the Products and it also covers if the Products do not conform to the Product specification, excluding minor defects, if reasonably acceptable and that do not compromise efficiency in their use.

The warranty does not apply to any Product (including any component or other parts in such Products (such as suction cups, filter elements, sealings, hoses, foam, etc.) or to the software of any Products) that it was used other than the intended purpose, and: (a) has been subjected to abuse, misuse, negligence, improper storage, improper handling, improper use, improper installation, abnormal physical stress, abnormal environmental or working conditions, or use, application, installation, care, control or maintenance contrary to any applicable manual or instructions for the Products issued by the Seller or good trade practice regarding the same; or (b) has been reconstructed, repaired or altered by any persons or entities other than the Seller or its authorized representatives, or have a defect as a result of fair wear and tear or willful damage or caused by subsequent damages caused by other defective products.

The product warranty set forth in this Section is the only warranty given by the Seller in relation to the Products. The Customer may not rely and has not relied, on any other information, statement, or warranty (express or implied), whether based on applicable law or otherwise. In any case, the compensation is limited to the price of the products agreed between the parties and is excluded from indirect damages.

During the warranty period, the Seller shall replace or repair, at its own expense, faulty products determined by the Seller, in its sole discretion, to be covered by the warranty set out herein.

It is at the Seller's discretion whether a faulty Product should be returned to the Seller for replacement or if it should be repaired by the Seller at the location of the Customer. Any replaced Products shall become the property of the Seller.

The Seller is not responsible for the cost of fitting replacement parts or components of any Products into any products or alike of the Customer.

These Terms and Conditions shall apply to any repaired or replaced Products by the Seller.

11. Recycling and disposal



Environmental aspects are considered in the development process of Piab's products to make sure that a minimal environmental footprint is used.

Piab AB is certified with ISO-14001:2015.

Piab AB also complies with:

- RoHS2 (2011/65/EU)
- REACH (EC 1907/2006)

The ways of handling recycling and disposal vary from country to country, and therefore this process needs to be in full compliance with each national regulation. If possible, disassemble the product into its various components. Electrical and electronic equipment should be handed over to an authorized body for disposal, as well as the metal parts. All other parts can either be recycled or sorted as waste.

For further information regarding REACH, visit <https://www.piab.com/downloads/document-center-page>



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